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BARKER COLLEGE

FORM VI

MATHEMATICS

2/3 UNIT

AM THURSDAY 17 AUGUST

H.S.C. TRIAL 1989.

TIME: 3 HOURS

200 copies

INSTRUCTIONS:

- * ALL questions may be attempted, and are of equal value.
- * All necessary working should be shown in every question. Marks may not be awarded for careless or badly arranged work.
- * Standard integrals are printed at the end of the paper.
- * Approved calculators may be used.
- * Each question attempted is to be returned in a separate Booklet marked Question 1, Question 2, etc on the cover. Each Booklet must show your Candidate's Number.
- * If you do not attempt a question, you must still hand in a Booklet for that question with NOT ATTEMPTED written clearly on the front.

* * * * *

Question 1:

(a) Evaluate: $\frac{\sqrt{16 \cdot 04}}{1.2 \times 3.56}$ correct to 2 decimal places.

(b) Solve the following equations:

(i) $y^2 - y - 12 = 0$

(ii) $\frac{x+1}{2} - \frac{x-1}{3} = 10$

(iii) $9^x = \frac{1}{27}$

(c) Factorise and simplify: $\frac{4m^2 - 4}{m - 1}$

(d) Rationalise the denominator and simplify: $\frac{\sqrt{2}}{\sqrt{2} - 1}$

(e) Given that $V = \frac{4}{3}\pi R^3$, find the value of R if $V = 1000$.
Give your answer correct to one decimal place.

Question 2:

(a) The points A and B have coordinates (-3,6) and (5,2) respectively.
Plot these points on a number plane.

(i) Find the equation of the line l joining the points A and B.

(ii) The line k is drawn through B with a gradient of 2. Show that the equation of k is:
 $2x - y - 8 = 0$

(iii) Show that the line k passes through P(2,-4).

(iv) Prove that the triangle ABP is right-angled.

(v) Find the area of triangle ABP.

(b) Sketch the graph of $y = \sqrt{9 - x^2}$. State its domain and range.

Question 3:

(a)

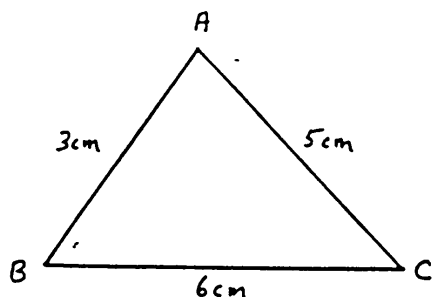


figure not to scale

- (i) Use the Cosine rule to calculate $\angle BAC$ to the nearest degree.
- (ii) Calculate the area of the triangle ABC. Give your answer correct to two significant figures.
- (b) An aircraft flies 500 km from its base on a bearing of 050° . It then flies 240 km due North. Calculate its distance (nearest km) from its base and the bearing (nearest degree) in which it must fly to return to its base in a direct line.
- (c) Prove that: $\tan A \sin A + \cos A = \sec A$
- (d) Sketch (not on graph paper): $y = 3 \sin 2x$ for $0 \leq x \leq 2\pi$

Question 4:

(a)

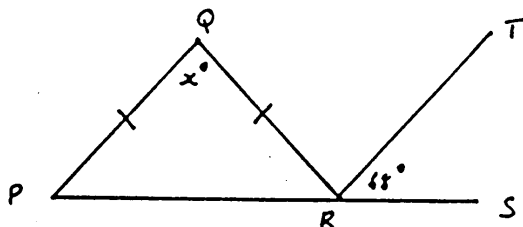
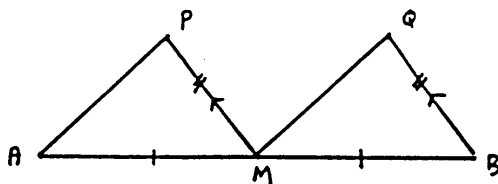


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Given that RT is parallel to PQ and that triangle PQR is isosceles, find the value of x , giving reasons.

(b)



M is the midpoint of AB, $PM = BQ$ and $PM \parallel BQ$

- (i) draw a neat copy in your exam book.
- (ii) Prove that $\triangle PAM \cong \triangle QMB$
- (iii) Prove: $AP \parallel MQ$

QUESTION 4 C'tinued:

(c)

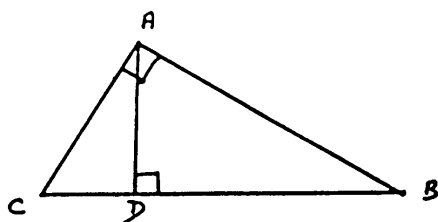


figure not to scale.

ABC is a triangle, right-angled at A and AD is drawn perpendicular to BC.

AB = 15 cm and AD = 12 cm

- (i) draw a neat sketch and mark on it all the given information
- (ii) calculate the length of BD
- (iii) PROVE $\triangle ABC$ is similar to $\triangle DBA$
- (iv) find the length of AC

Question 5:

(a) Differentiate with respect to x:

(i) $y = 4x^3 - 2x + \frac{1}{x^2} - 1$

(ii) $y = x^2 e^x$

(iii) $y = \frac{x+1}{x-1}$

(b) Evaluate the following integrals:

(i) $\int_1^4 \left(x - \frac{1}{x^2}\right) dx$

(ii) $\int_0^1 (2x - 1)^4 dx$

(iii) $\int_0^{\frac{\pi}{4}} \cos 2x dx$

Question 6:

(a) Without solving the equation $2x^2 - 3x + 6 = 0$ which has the roots α and β , find the value of:

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\alpha^2 + \beta^2$

(b) For what values of k does the equation $x^2 - (k+1)x + 1 = 0$ have:

(i) equal roots

(ii) no real solutions

(c) Given that the parabola has the focus at (2,1) and vertex at (2,3) find:

(i) focal length

(ii) directrix

(iii) axis of symmetry

(iv) equation of the parabola

(v) sketch the parabola

Question 7:

- (a) Which term of the sequence 2, 6, 18 is 486?
- (b) A contractor undertakes to bore a well for \$9 for the first 10 metres, \$10 for the next 10 metres, \$11 for the next 10 metres and so on. What is the cost of a well 500 metres deep?
- (c) Given: $f(x) = x^3 - 12x$
- (i) find the coordinates of the maximum and minimum turning points
- (ii) find the coordinates of any points of inflexion (if they exist)
- (iii) DRAW a neat sketch, indicating all essential features.

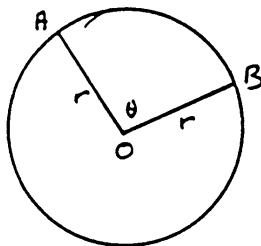
Question 8:

- (a) Evaluate the $\lim_{x \rightarrow \infty} \frac{x^2 - 2x + 5}{2x^2 + 3x - 7}$
- (b) Find the equation of the curve $y = f(x)$ given that $\frac{dy}{dx} = 1 - 5x^4$ and that the curve passes through (1,3)
- (c) Sketch the curves $y = x^2 - 1$ and $y = 1 - x^2$ on the SAME number plane. Hence find the area enclosed between the two curves.
- (d) Calculate the area enclosed between the curve $y = \frac{1}{4x + 1}$, the x axis, $x = 1$ and $x = 5$
- (i) by Simpson's rule using 3 function values
- (ii) by integration (the exact value).

Question 9:

- (a) Solve the equation: $2 \cos x + 1 = 0$ for $0 \leq x \leq 2\pi$
- (b) Find the equation of the tangent to the curve $y = \sin 2x$ when $x = \frac{\pi}{2}$

(c)



In a circle radius r , the arc AB subtends an angle θ at the centre.

- (i) Find an expression for the length of arc AB and hence write down an expression for the perimeter of the sector AOB
- (ii) If the perimeter of the sector AOB is 8 cm, write an expression for θ in terms of r
- (iii) Find the maximum area of the sector AOB when the perimeter of the sector AOB is 8 cm.

Question 10:

- (a) Find the volume of the solid of revolution formed by rotating the curve $y = e^x + e^{-x}$ about the x axis between $x = -1$ and $x = 1$.
- (b) On the birth of their daughter Karen, the proud parents Richard and Helen decided to start an investment fund for her. They agreed to deposit \$500 on the 1st August 1989 (the date of birth) and \$500 on every subsequent birthday up to and including her 21st birthday. If we assume the interest is being paid at the rate of 15% p.a. (compound interest), find how much Karen will receive on her 21st birthday?
- (c) The temperature of a cup of black coffee is given by $T = 100e^{-t/5}$ where t is the time in minutes.
If it is too hot to drink above 55°C and too cold below 25°C , calculate the length of time during which the coffee is drinkable (to the nearest second).

END OF PAPER

Unit Solutions (Trials)

1. a) 0.14 2 d.p. ①

b) i) $(y-4)(y+3) = 0$
 $y = 4, -3$ ①

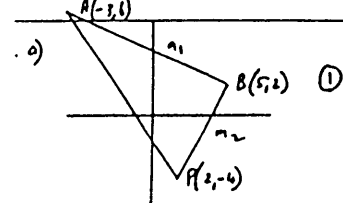
ii) $3(x+1) - 2(x-1) = 60$
 $x+5 = 60$
 $x = 55$ ①

iii) $3^{2x} = 3^{-3}$ ① $\Rightarrow x = -\frac{3}{2}$ ①

c) $\frac{4(m-1)}{m-1} = \frac{4(m-1)(m+1)}{(m-1)}$ ① $= 4(m+1)$ ①

d) $\frac{\sqrt{2}(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)}$ ① $= \frac{2+\sqrt{2}}{2-1} = 2+\sqrt{2}$ ①

e) $R = \sqrt[3]{\frac{1000}{4\pi}}$ ① $= 6.2035$
 $= 6.2$ ① 1 d.p.



i) eqn AB $m_1 = \frac{6-2}{-3-5} = -\frac{4}{8} = -\frac{1}{2}$ ①
 $y-6 = -\frac{1}{2}(x+3)$
 $2y-12 = -x-3$
 $x+2y-9=0$ ①

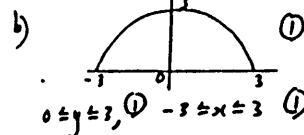
ii) $A=2$, $(5,2)$ $y-2 = 2(x-5)$ ①
 $2x-y-8=0$

iii) if $(2,-6)$ lies on line: must satisfy the eqn. $4+(-6)-8=0$ YES ①

iv) since $m_1 = -\frac{1}{2}$ & that $m_2 = 2$ the gradient of line k & that $m_1 m_2 = -1$ ①
 $\therefore \triangle ARP$ is rt. angled

v) area $= \frac{1}{2} BP \cdot AR$
 $BP = \sqrt{2^2+6^2} = \sqrt{40} = 2\sqrt{10}$ ①
 $AR = \sqrt{4^2+4^2} = \sqrt{32} = 4\sqrt{2}$ ①

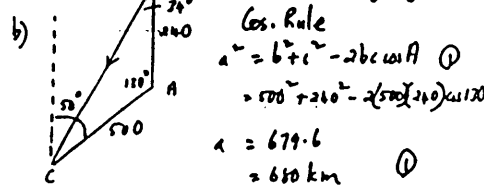
$\therefore \text{area} = \frac{1}{2} 2\sqrt{10} \cdot 4\sqrt{2} = 30 \text{ unit}^2$ ①



Q3. a) i) $\cos A = \frac{b^2+c^2-a^2}{2bc}$ ① $= \frac{5^2+3^2-6^2}{2 \cdot 5 \cdot 3} = -\frac{1}{5}$ ①

$\angle BAC > 90^\circ$ nearest deg. ①

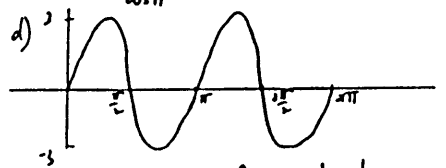
ii) area $\triangle BAC = \frac{1}{2} \times 3 \times 5 \sin 94^\circ$
 $= 7.5 \text{ cm}^2$ 2 sig. fig. ①



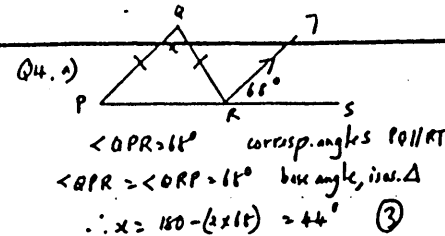
cos. Rule
 $a^2 = b^2 + c^2 - 2bc \cos A$ ①
 $600^2 = 500^2 + 700^2 - 2(500)(700) \cos 110^\circ$
 $a = 679.6$
 $= 680 \text{ km}$ ①

Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B}$ ①

$\frac{600}{\sin 110^\circ} = \frac{500}{\sin B}$ now $\sin 110^\circ = \sin 70^\circ$
 $\therefore \sin B = \frac{500 \sin 70^\circ}{600}$ ①
 $\angle B = 36^\circ$ ①

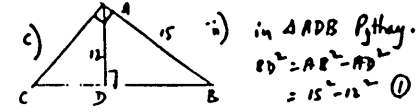


① for period & shape
 ② for amplitude



Q4. a) $\angle QPR = 68^\circ$ corresp. angles $PQ \parallel RT$
 $\angle QPR = \angle QRP = 68^\circ$ base angle, isos. $\triangle$
 $\therefore x = 180 - (2 \times 68) = 44^\circ$ ③

b) $\angle PMA = \angle QMB$ corresp. $\angle$ s $PM \parallel QB$
 $PM = QM$ given
 $AM = MB$ given
 $\therefore \triangle PMQ \equiv \triangle QMB$ (S.A.S.) ③
 $\therefore \angle PMQ = \angle QMB$ corresp. $\angle$ s in congruent $\triangle$ s
 since corresp. $\angle$ s equal $\therefore PM \parallel QB$ ①



c) in $\triangle ADB$ Pythag.
 $BD^2 = AB^2 - AD^2$
 $= 15^2 - 12^2$ ①
 $BD = 9$

ii) $\angle RDA = \angle BAC = 90^\circ$ given ②
 $\angle ABD = \angle CBA$ common angle
 $\therefore \triangle ABC$ is similar to $\triangle DBA$
 iv) let $AC = x$ $\frac{4}{15} = \frac{x}{20}$ $x = \frac{120}{5} = 24$ ②

Q5. a) i) $y' = 12x^2 - 2 - 2x^{-1}$ ②
 ii) $y' = 2x e^x + x^2 e^x$ ②
 $= x e^x (2 + x)$
 iii) $y' = \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2} = \frac{-2}{(x-1)^2}$ ②

i) $\int_1^4 (x-x^{-1}) dx = \left[\frac{1}{2} x^2 + x^{-1} \right]_1^4$
 $= \left(8 + \frac{1}{4} \right) - \left(\frac{1}{2} + 1 \right)$ ②
 $= \frac{6\frac{1}{4}}{1}$

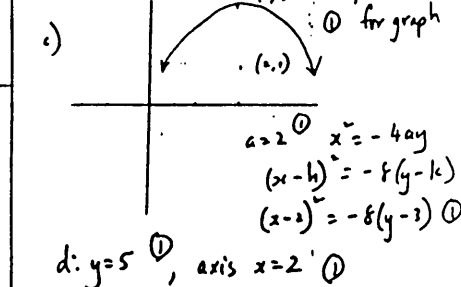
ii) $\int_0^1 (2x-1)^5 dx = \left[\frac{(2x-1)^6}{2 \cdot 6} \right]_0^1$ ②
 $= \frac{1}{10} - \left(-\frac{1}{10} \right) = \frac{2}{10} = \frac{1}{5}$

iii) $\int_0^{\frac{\pi}{2}} \cos 2x dx = \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$
 $= \frac{1}{2} \sin \pi - \frac{1}{2} \sin 0$ ②
 $= \frac{1}{2}$

Q6. a) $2x^2 - 3x + 1 = 0$
 i) $\alpha + \beta = -\frac{-3}{2} = \frac{3}{2}$ ① ii) $\alpha \beta = \frac{1}{2} = 3$ ①
 iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ ①
 $= \left(\frac{3}{2} \right)^2 - 2 \cdot 3 = \frac{9}{4} - 6 = -\frac{3}{4}$ ①

b) i) $\Delta = 0$ for equal roots
 $b^2 - 4ac = (k+1)^2 - 4(1)(1) = 0$ ①
 $k^2 + 2k + 1 - 4 = 0$
 $k^2 + 2k - 3 = 0$
 $(k+3)(k-1) = 0$
 $k = -3, 1$ ①

ii) $\Delta < 0$ for no real solutions
 ie. $(k+3)(k-1) < 0$
 $-3 < k < 1$ ①



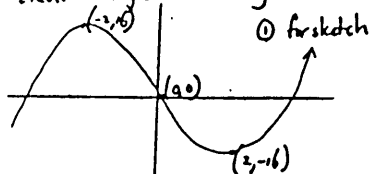
d: $y = 5$ ①, axis $x = 2$ ①

Q7. a) $a=2, r=3$
 $T_r = ar^{n-1} = 486$ ①
 $2 \cdot 3^{n-1} = 486$
 $3^{n-1} = 243 = 3^5$ ①
 $n-1 = 5$ ①
 $n = 6$ ①

b) $a=50, d=9, n=1$ ①
 $S_n = \frac{n}{2} [2a + (n-1)d]$ ①
 $= 25 (18 + 49) = 816.75$ ①

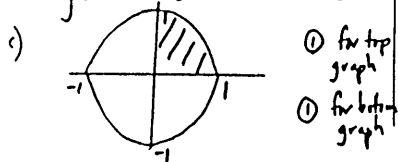
c) $y = x^3 - 12x$
 $y' = 3x^2 - 12 = 3(x^2 - 4)$
 $3(x-2)(x+2) = 0$
 $x = 2, -2$ (2, -16) (-2, 16)
 $y'' = 6x$, when $x=2, y'' > 0$ (2, -16) min
 $x=-2, y'' < 0$ (-2, 16) max

① at $x=0, y''=0$ (0,0) pt. of inf.
 ① check - change in concavity

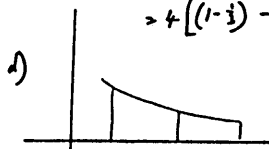


Q8. a) $\lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x} + \frac{1}{x^2}}{2 + \frac{1}{x} - \frac{2}{x^2}} = \frac{1}{2}$ ①

b) $y' = 1 - 5x^4$
 $y = x - x^5 + c_1$
 $3 = 1 - 1 + c_1$ (c₁ = 3)
 $y = x - x^5 + 3$ ①



area enclosed $= 4 \int_0^1 (1-x^2) dx$ ①
 $= 4 \left[x - \frac{1}{3}x^3 \right]_0^1$
 $= 4 \left[(1 - \frac{1}{3}) - 0 \right] = \frac{8}{3}$ ①



d) i) Simpson's Rule $A \approx \frac{1}{3} [f(a) + f(b) + 4f(\frac{a+b}{2})]$ ①
 $= \frac{1}{3} [\frac{1}{5} + \frac{1}{5} + 4 \cdot \frac{1}{5}]$ ①
 $h = \frac{b-a}{n} = \frac{5-1}{2} = 2$
 ii) $A = \int_1^5 \frac{1}{4x+1} dx = \int_1^5 \frac{1}{4x+1} dx = \frac{1}{4} \int_1^5 \frac{4}{4x+1} dx$ ①
 $= \frac{1}{4} [\log(4x+1)]_1^5$
 $= \frac{1}{4} [\log 21 - \log 5] = \frac{1}{4} \log(\frac{21}{5})$ ①

Q9. a) $2 \cos x + 1 = 0, 4 \sin x = -\frac{1}{2}$ ①
 $x = \frac{2\pi}{3}, \frac{5\pi}{3}$

b) $y' = 2 \cos 2x$ at $(\frac{\pi}{2}, 0)$ ①
 $m = 2 \cos \pi = 2(-1) = -2$
 $y - 0 = -2(x - \frac{\pi}{2})$ ①
 $2x + y - \pi = 0$

c) arc AB $= r\theta$ ①

let $P = 2r + r\theta$
 now $P = 8 = 2r + r\theta$
 $r\theta = 8 - 2r, \theta = \frac{8}{r} - 2$ ①

area sector $A = \frac{1}{2} r^2 \theta$ ①
 $= \frac{1}{2} r^2 (\frac{8}{r} - 2)$
 $= 4r - r^2$ ①
 $\frac{dA}{dr} = 4 - 2r = 0$ ①
 $r = 2$

$\frac{dA}{dr} = 4 - 2r < 0 \Rightarrow \text{MAX}$ ①
 $\therefore \text{max area} = \frac{1}{2} r^2 \theta$ when $r=2$ ①
 $= \frac{1}{2} (4) (2)$
 $= 4 \text{ cm}^2$ ①

Q10 a) $V = \int_a^b \pi y^2 dx$ around x-axis ①
 $= \int_{-1}^1 \pi (e^x + e^{-x})^2 dx = 2 \int_0^1 \pi (e^x + e^{-x})^2 dx$
 since even fn.
 $= 2\pi \int_0^1 (e^{2x} + 2e^x e^{-x} + e^{-2x}) dx$ ①
 $= 2\pi \int_0^1 (e^{2x} + 2 + e^{-2x}) dx$
 $= 2\pi \left[\frac{1}{2} e^{2x} + 2x - \frac{1}{2} e^{-2x} \right]_0^1$ ①
 $= 2\pi \left[\left(\frac{1}{2} e^2 + 2 - \frac{1}{2} e^{-2} \right) - \left(\frac{1}{2} + 0 - \frac{1}{2} \right) \right]$
 $= 2\pi \left(\frac{1}{2} e^2 + 2 - \frac{1}{2} e^{-2} - \frac{1}{2} + \frac{1}{2} \right)$
 $= 2\pi \left(\frac{1}{2} e^2 + 2 - \frac{1}{2} e^{-2} \right)$ ①

b) $A = P(1 + \frac{r}{100})^n$ ①
 $A = 500(1.15)^n + 500(1.15)^{n-1} + \dots + 500(1.15)^0$
 at birthday - no interest ①
 $= 500 [1 + 1.15 + 1.15^2 + \dots + 1.15^n]$
 $= 500 \frac{a(r^n - 1)}{r - 1} = 500 \frac{1(1.15^n - 1)}{1.15 - 1}$ ①
 $= 500 \frac{(21.66746 - 1)}{0.15}$
 $= 168815.42$ ①

c) $T = 100e^{-\frac{t}{5}}$
 $T = 55 = 100e^{-\frac{t}{5}}$
 $0.55 = e^{-\frac{t}{5}}$
 $\log(0.55) = -\frac{t}{5}$
 $t = 2.919 \text{ min}$ ②

$T = 25 = 100e^{-\frac{t}{5}}$
 $0.25 = e^{-\frac{t}{5}}$
 $t = 6.4314$ ②
 change in time = 3.9 min
 $= 7 \text{ min } 56 \text{ s (nearest sec.)}$